Satwik Mishra

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Education

Indian Institute of Information Technology, Tiruchirappalli (IIIT Trichy) **Nov 2020 – Apr 2024**
B.Tech, Electronics and Communication Engineering | Institute of National Importance *Tiruchirappalli, India*

Technical Skills

Languages: JavaScript, TypeScript, C++, Java, SQL
Backend: Node.js, Spring Boot, REST APIs, OAuth 2.0/JWT, RBAC, Puppeteer
Frontend: React, Tailwind, IndexedDB
Data: Oracle (PL/SQL), PostgreSQL (Supabase), Cloudflare D1
Cloud & DevOps: Cloudflare (Workers, Pages, Access), GitHub Actions CI/CD

Projects

Career Page Monitor **2026**
Node.js, Puppeteer, PostgreSQL (Supabase), React, Cloudflare Workers, GitHub Actions GitHub

- Built a scraping system for **120+ company career pages**, with shared adapters for **6 ATS platforms** (Workday, Greenhouse, Lever, Ashby, SmartRecruiters, Personio) and **headless-browser** scrapers for custom sites.
- Orchestrated **hourly runs** via a **Cloudflare Worker cron** dispatching **GitHub Actions** workflows, after GitHub's native scheduler ran only 3–4 times a day under load.
- Made runs **idempotent** with a **unique constraint** on (company, job ID) in **PostgreSQL**, sending **Telegram alerts** only for new openings so referrals can be requested within the hour.
- Kept the database inside **Supabase's 500 MB free tier** by aging out postings on a **7-day / 30-day lifecycle**, so the system runs at **zero hosting cost**.
- Built a **React** dashboard on **Cloudflare Pages** behind **Cloudflare Access**, tracking each job from interested to applied, with a **scraper-health panel** flagging timed-out or failing sources.

Chunky Drive — Chunked File Storage **2024 – 2026**
React, TypeScript, Cloudflare Workers, D1 GitHub | Live

- Built a Google-Drive-style app on an external storage backend: the browser splits files into **8 MB chunks** and hashes each with **SHA-256**, deriving the file hash from the chunk hashes and verifying both on reassembly.
- Served the app from a **Cloudflare Worker** that holds storage credentials and **streams chunks without buffering** to stay within its **10 ms CPU limit**, moving **3 in parallel** with **retry and backoff**; **D1** stores only the chunk map (∼25 MB per TB).
- Made **downloads resumable** across closed tabs by persisting verified chunks in **IndexedDB**, with **Google OAuth + email OTP** auth and a **cron** cleaning up abandoned uploads.

Experience

Tata Consultancy Services **Nov 2024 – Present**
Systems Engineer — Innovator (TCS's top engineering hiring track)

- Co-built **PDN Tracker**, an internal **deployment-management platform** (React, **Spring Boot**, **Oracle PL/SQL**) replacing a deprecated vendor tool and serving **500+ daily active users**; owned the **frontend and middleware**.
- Modeled the release lifecycle (draft, review, testing, delivery, release) as a **table-driven state machine** mapping each role and status to its allowed actions, **validated server-side**, with send-back reasons and rejected states kept for **audit history**.
- Replaced manual role selection at login with **session-based auth** and **role-embedded access tokens (RBAC)**, eliminating wrong-role login failures and cutting total login errors by **35%**.
- Migrated **10+ legacy SOAP services** on **TCS BaNCS** (core banking platform used by banks worldwide) to **REST APIs**, tracing each transaction's **call chain and parameter derivation** across UI, middleware, and **PL/SQL stored procedures** in a **30+ GB monolithic codebase** to define **API contracts** with behavioral parity.
- Cut a policy-issuance API from **5s+ (timing out) to 250 ms** by **parallelizing 5 sequential DB calls** on a **Java thread pool** and adding **targeted indexes**, with **compensating cleanup** if any call failed.
- Performed **root-cause analysis** on production incidents across application logs, Oracle data, and PL/SQL procedures, resolving the **data-integrity** issues behind the majority of tickets.

Achievements & Activities

LeetCode Knight (2023, rating **1882**) — 500+ problems solved
Coordinator, Coding Club & Web Dev Club, IIIT Trichy — Organized coding contests, workshops, and hackathons.
Football — Midfielder for the CBSE cluster team, competed at the **state level**.